UNIVERSALLY CLOSED SURJECTION FROM AN AFFINE SPACE HAS AFFINE TARGET

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 3649-3692, following the statement at 3640-3647.

Editorial note. The footnote (integral morphisms are universally closed) and the one-line 'X = lim X_i' via relative Spec colimit compatibility are transcribed.

Allowed standard prerequisites (editorial).

- Image of universally closed from separated source. Morphisms of Spaces, Lemma spaces-morphisms-lemma-image-universally-closed-separated (cross-chapter, cited at ~3653-3655). Y is separated.
- Affine permanence. Morphisms of Spaces, Lemma spaces-morphisms-lemma-affine-permanence (cross-chapter, cited at ~3656-3657). f is affine.
- Integral morphisms and universal closedness. Morphisms of Spaces, Lemma spaces-morphisms-lemma-integral-universally-closed (cross-chapter, cited at ~3658-3660 and footnote ~3644-3646). f is integral; the footnote notes integral implies universally closed.
- Pushforward of quasi-coherent algebras. Morphisms of Spaces, Lemma spaces-morphisms-lemma-pushforward (cross-chapter, cited at ~3670-3671). A = f_*O_X is a quasi-coherent O_Y-algebra.
- Colimit of finite type modules. lemma-colimit-finitely-presented (same chapter, cited at ~3672-3674). A = colim F_i with F_i finite type.
- Relative Spec for quasi-coherent algebras. Morphisms of Spaces, Lemma spaces-morphisms-lemma-affine-equivalence-algebras (cross-chapter, cited at ~3683-3684). X_i = Spec_Y(A_i) is finite over Y.
- Affineness of the limit descends. lemma-limit-is-affine (same chapter, card L26293, cited at ~3686-3688). X_i affine for i large.
- Target of a finite surjection is affine. lemma-affine (same chapter, cited at ~3690-3691). Concludes Y is affine.

Source reference. Original lemma 07VS: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 3592–3592. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 07V9: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 2492–2492. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Date: stacks-project:spaces-limits.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 07SQ: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 1370–1370. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 03I5: focused-seeds/geometry/S9-spaces-limits-proposition-affine/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 081V: focused-seeds/geometry/S9-spaces-limits-proposition-affine/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 08GB: focused-seeds/geometry/S9-spaces-limits-proposition-affine/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 05Z2: focused-seeds/geometry/S9-spaces-limits-proposition-affine/ref-tags.txt, source lines 8–8. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0415: focused-seeds/geometry/S9-spaces-limits-proposition-affine/ref-tags.txt, source lines 9–9. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 03M9: focused-seeds/geometry/S9-spaces-limits-proposition-affine/ref-tags.txt, source lines 10–10. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Proposition 0.1. *Let S be a scheme. Let $f : X \to Y$ be a morphism of algebraic spaces over S. Assume that X is affine and f is surjective and universally closed*[1]*. Then Y is affine.*

Proof. We may and do view $f : X \to Y$ as a morphism of algebraic spaces over $\mathrm{Spec}(\mathbf{Z})$ (see Spaces, Definition 03I5). By Morphisms of Spaces, Lemma 05Z2 we see that Y is a separated algebraic space. Then by Morphisms of Spaces, Lemma 08GB we find that f is affine. Whereupon by Morphisms of Spaces, Lemma 0415 we see that f is integral.

[1] An integral morphism is universally closed, see Morphisms of Spaces, Lemma 0415.

By the preceding paragraph, we may assume $f : X \to Y$ is surjective and integral, X is affine, and Y is separated. Since f is surjective and X is quasi-compact we also deduce that Y is quasi-compact.

Consider the sheaf $\mathcal{A} = f_*\mathcal{O}_X$. This is a quasi-coherent sheaf of $\mathcal{O}_Y$-algebras, see Morphisms of Spaces, Lemma 03M9. By Lemma 07V9 we can write $\mathcal{A} = \operatorname{colim}_i \mathcal{F}_i$ as a filtered colimit of finite type $\mathcal{O}_Y$-modules. Let $\mathcal{A}_i \subset \mathcal{A}$ be the $\mathcal{O}_Y$-subalgebra generated by $\mathcal{F}_i$. Since the map of algebras $\mathcal{O}_Y \to \mathcal{A}$ is integral, we see that each $\mathcal{A}_i$ is a finite quasi-coherent $\mathcal{O}_Y$-algebra. Hence

$$X_i = \underline{\operatorname{Spec}}_Y(\mathcal{A}_i) \longrightarrow Y$$

is a finite morphism of algebraic spaces. Here $\underline{\operatorname{Spec}}$ is the construction of Morphisms of Spaces, Lemma 081V. It is clear that $X = \overline{\lim}_i X_i$. Hence by Lemma 07SQ we see that for i sufficiently large the scheme X_i is affine. Moreover, since $X \to Y$ factors through each X_i we see that $X_i \to Y$ is surjective. Hence we conclude that Y is affine by Lemma 07VS. $\square$